

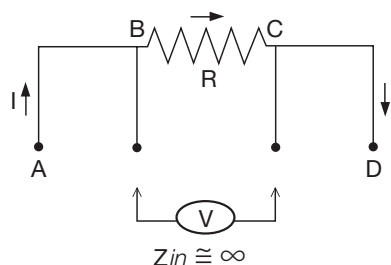
**Ultra High Precision Z-Foil Surface Mount
Current Sensing Chip Resistor**
with Power Rating to 1 W and Temperature Coefficient of Resistance
of ± 0.05 ppm/°C and Power Coefficient of 5 ppm at Rated Power

FEATURES

- ± 0.05 ppm/°C typical (0°C to +60°C)
 ± 0.2 ppm/°C typical (–55°C to +125°C, +25°C ref.)
- Resistance range: 0.3 Ω to 10 Ω
- Resistance Tolerance: to $\pm 0.2\%$
- Power coefficient “ ΔR due to self heating”:
5 ppm at rated power
- Load life stability: 0.015% at 70°C,
2000 h at rated power
- Electrostatic discharge (ESD) up to 25 000 V
- Short time overload <0.005%



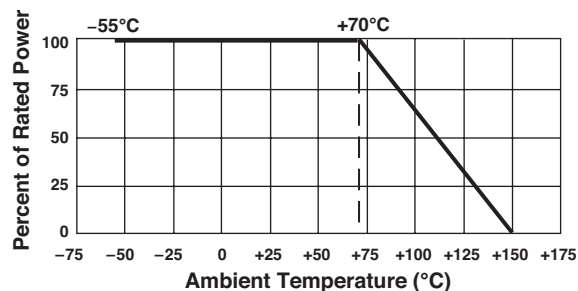
RoHS*
COMPLIANT

**Specifications⁽¹⁾**

MODEL NUMBER	RESISTANCE RANGE	RESISTANCE TOLERANCE ⁽⁴⁾	TYPICAL TCR and MAX. SPREAD (–55°C to +125°C, +25°C Ref)	POWER RATING at +70°C ⁽²⁾	MAXIMUM CURRENT ⁽²⁾
VCS1625ZP	>2.0 Ω to 10 Ω 0.3 Ω to 2.0 Ω	$\pm 0.2\%$, $\pm 0.5\%$; $\pm 1.0\%$ $\pm 0.5\%$; $\pm 1.0\%$	$\pm 0.2 \pm 2.8$ ppm/°C	1 W on FR4 PCB ⁽³⁾	1.8 A

Notes

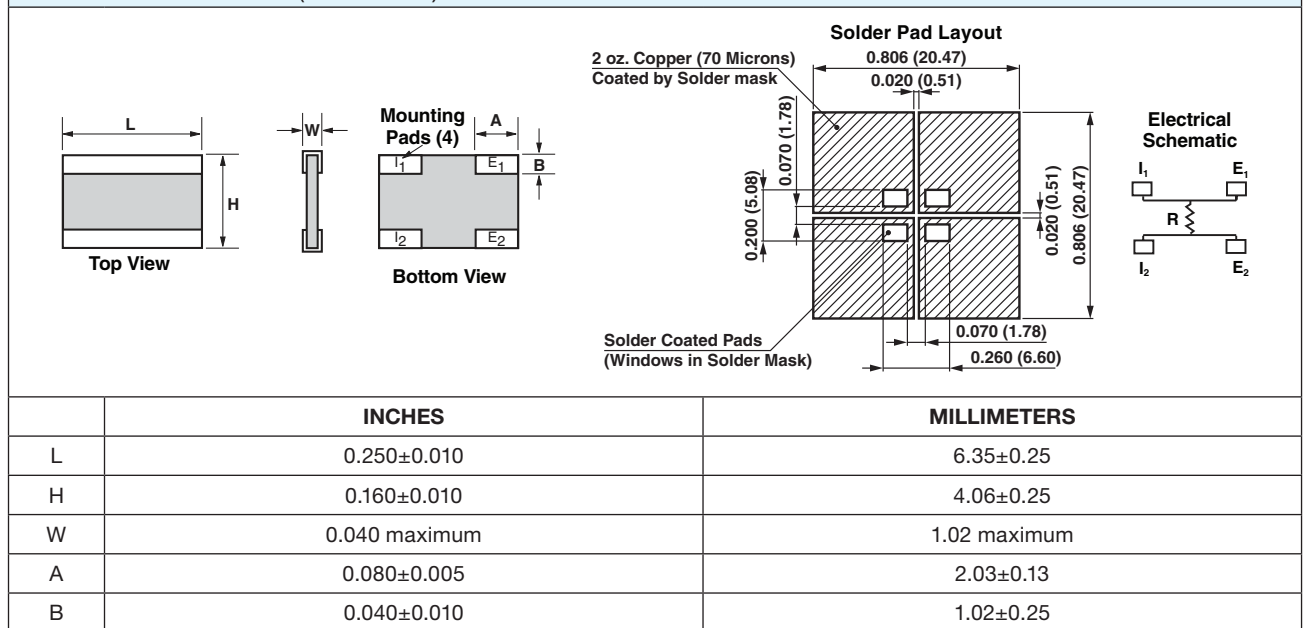
- ⁽¹⁾ Tighter performances are available. Please contact application engineering: foil@vpgsensors.com
⁽²⁾ Whichever is lower
⁽³⁾ See solder pad layout
⁽⁴⁾ 0.05% and 0.1% are also available per special request

Power Derating Curve**Note**

* This datasheet provides information about parts that are RoHS-compliant and/or parts that are non-RoHS-compliant. For example, parts with lead (Pb) terminations are not RoHS compliant.

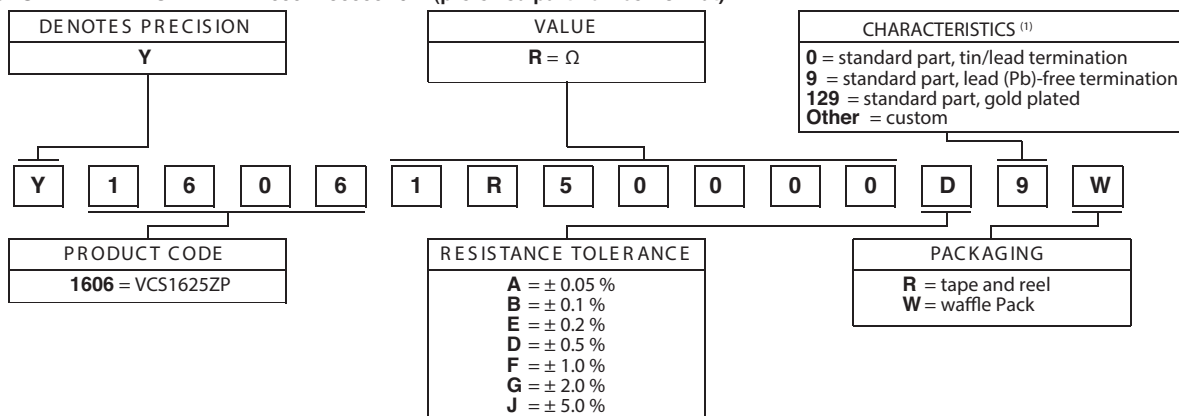
Performance Specifications

TEST	MIL-PRF-55342 ΔR LIMITS	TYPICAL ΔR LIMITS	MAXIMUM ΔR LIMITS ⁽¹⁾
Thermal shock, $5 \times (-65^{\circ}\text{C to } +150^{\circ}\text{C})$	$\pm 0.10\%$	$\pm 0.005\%$ (50 ppm)	$\pm 0.01\%$ (100 ppm)
Low temperature operation, -65°C , 45 min at P_{nom}	$\pm 0.10\%$	$\pm 0.005\%$ (50 ppm)	$\pm 0.01\%$ (100 ppm)
Short time overload, $6.25 \times$ rated power, 5 s	$\pm 0.10\%$	$\pm 0.005\%$ (50 ppm)	$\pm 0.02\%$ (200 ppm)
High temperature exposure, $+150^{\circ}\text{C}$, 100 h	$\pm 0.10\%$	$\pm 0.01\%$ (100 ppm)	$\pm 0.02\%$ (200 ppm)
Resistance to soldering heat	$\pm 0.2\%$	$\pm 0.01\%$ (100 ppm)	$\pm 0.03\%$ (300 ppm)
Moisture resistance	$\pm 0.2\%$	$\pm 0.01\%$ (100 ppm)	$\pm 0.03\%$ (300 ppm)
Load life stability, 2000 h at 70°C at 1 W	$\pm 0.5\%$	$\pm 0.015\%$ (150 ppm)	$\pm 0.025\%$ (250 ppm)

Note⁽¹⁾ Measurement error 0.001R**Dimensions** in inches (millimeters)

Global Part Number Information

NEW GLOBAL PART NUMBER: Y16061R50000D9W (preferred part number format)



FOR EXAMPLE: ABOVE GLOBAL ORDER Y1606 1R50000 D 9 W:

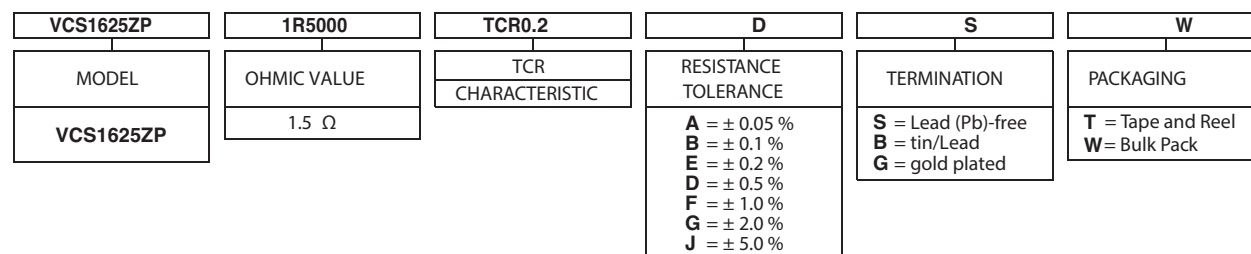
TYPE: VCS1625ZP

VALUE: 1.5 Ω ABSOLUTE TOLERANCE: $\pm 0.5\%$

TERMINATION: lead (Pb)-free

PACKAGING: waffle pack

HISTORICAL PART NUMBER: VCS1625ZP 1R5000 TCR0.2 D S W (will continue to be used)



Note

⁽¹⁾ For non-standard requests, please contact application engineering.